International Journal of Sports Physiology and Performance, (Ahead of Print)
https://doi.org/10.1123/ijspp.2023-0033
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First Published Online: June 13, 2023

Human Kinetics
BRIEF REPORT

Thinking About Elite Performance: The Experience and Impact of Mental Fatigue in Elite Sport Coaching

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Purpose: Mental fatigue causes decreases in aspects of athletes' performance. Elite coaches commonly undertake cognitively demanding tasks and are seemingly at similar risk of subsequent performance impairment. However, elite sport coaches' experiences of mental fatigue, alongside other markers of psychobiological stress, have yet to be quantified. ***Methods:*** Three elite coaching and performance staff (2 women and 1 man) provided 100-mm visual analog scale ratings of mental fatigue, physical fatigue, readiness to perform, and salivary samples for later cortisol (sCort) and alpha-amylase (sAA) analysis. Data were obtained on the same morning each week across a 16-week preseason. Data were subset by individual coach for descriptive and repeated-measures correlational analyses. ***Results:*** Fluctuating mental fatigue was observed over the 16 weeks (min–max; coach 1 = 25–86 AU; coach 2 = 0–51 AU; and coach 3 = 15 – 76 AU). Elevated levels of mental fatigue were reported at multiple time points, with individual variability observed. sCort (in nanomoles per liter), sAA (in micromoles per liter), and sAA:sCort indicated that coaches experienced psychophysiological stress (min–max; coach 1 sCort = 8.42–17.31, sAA = 52.40–113.06, sAA:sCort = 3.20–12.80; coach 2 sCort = 4.20–9.70, sAA = 158.80–307.20, sAA:sCort = 21.10–61.70; and coach 3 sCort = 6.81–19.66, sAA = 86.55–495.85, sAA:sCort = 4.90–35.50). A significant inverse relationship between mental fatigue and readiness to perform ($r = -.44$ [−.64 to −.17], $P = .002$) was identified. ***Conclusions:*** Elite sport coaches report elevated instances of mental fatigue during a preseason training period. Those involved in elite sports should act to understand the presence and potential subsequent impacts of staff mental fatigue and consider management or mitigation strategies. Optimization of the cognitive performance of coaches and performance staff presents as a potential source of competitive advantage.

Keywords: cognition, mental performance, coaching performance, psychobiological, longitudinal

Increased scientific attention has recently been directed toward exploring mental fatigue in elite sport, with impairments in physical, technical, tactical, and psychological aspects of athlete performance evidenced.[1] However, optimization of elite sporting performance requires effective multidisciplinary management of athletes by coaching and performance staff. Effective coaching involves complex intellectual and interpersonal practices, deployed within ever-changing, interdependent environments. Mental fatigue is a potential consequence of prolonged demanding cognitive activity requiring sustained mental efficiency.[2] It is considered to be a psychobiological state, typically indicated by a change in subjective, behavioral, or (neuro)physiological markers.[3] Elite coaches and performance staff undertake tasks demanding information processing, working memory, response inhibition, information recall, sustained attention, and self-regulation.[4] One example of a task that requires such demands is coaching and the provision of live feedback during a game. This requires observation, interpretation of statistical information, reflection on intended tactics and current execution, and awareness of game momentum to call a time-out to provide feedback and instructions to athletes in a critical moment during the game, all while competing with crowd noise and visual stimuli.[4] Prior qualitative work has demonstrated that athletes and staff perceive mental fatigue to be induced by repetitive or long-duration activities, overanalysis, a demand for regular problem solving, the need to constantly think about the sport in question, unfamiliarity with an environment, and other commitments (eg, media and sponsorship engagements, contract negotiation, travel, relationship dynamics, and further study).[5]

The challenges faced and commitments of the coaches and sports science and sports medicine staff in supporting elite athletes are widely acknowledged.[6] However, the extent to which mental fatigue may influence the performance of coaching and support staff is yet to be examined. The impact of cognitive fatigue in other human performance domains, such as transportation and military settings, includes an increased shifting of attention to irrelevant stimuli and reduction in goal-directed attention,[7] impaired reaction

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time,[7] compromised executive control,[8] and impaired emotional regulation.[9] Accordingly, the potential presence of mental fatigue, and its subsequent consequences on the quality of service provision and athlete interactions, has the potential to negatively influence the preparation and performance of elite athletes. The primary purpose of this case study was to examine whether coaches experience mental fatigue, alongside other markers of psychobiological stress, during a preseason training phase. Second, the project aimed to explore the potential relationship between coach perceived mental fatigue and readiness to perform.

Methods

Subjects

Coaching and performance staff (n = 3) from an Australian professional Suncorp Super Netball team provided written informed consent to participate in the study. The staff consisted of one head coach, one assistant coach, and one head strength and conditioning coach (2 women and 1 man). The Suncorp Super Netball League is Australia's premier professional competition and is regarded as the best competition globally, attracting the most elite full-time athletes and coaches. Ethical approval was granted by an institutional human research ethics committee.

Methodology

Data were generated during a 16-week preseason (48 samples). During a typical week, the staff coached 4 full-team on-court training sessions (~120 min, including ~30 min of "conditioning" work), 3 resistance training sessions (~70 min, including key upper- and lower-body lifts), one analysis/tactical session (~60 min, minimal physical demand for players and no involvement of head strength and conditioning coach), and up to 3 specialist on-court skill sessions (~45 min), totaling ~15 hours of face-to-face coaching.[10] The overall coaching load was similar for each coach, with differing demands during specific sessions as per their job titles. The preseason phase was undertaken at the team's training base, including a preseason tournament during week 9. Staff were required to travel for preseason matches during weeks 7 and 12. Subjective ratings and a salivary sample were provided on the same morning each week. Subjective responses using 100-mm visual analog scales were obtained to indicate current levels of "physical fatigue," "mental fatigue" (anchored by "none at all = 0" to "maximal = 100"), and "readiness to perform" (anchored by "not at all ready = 0" to "maximally ready = 100"). "Readiness to perform" referred to how ready participants were to perform as a coach during the immediate upcoming session. Participants were familiarized with the visual analog scales and definitions, and salivary samples (Salivette, Sarstedt) were obtained and frozen for later analysis of salivary cortisol (sCort), alpha-amylase (sAA), and sAA:sCort, as per previously published protocols and coefficients of variation.[10]

Statistical Analysis

Data were analyzed using R statistical software (version 1.4.1717). Once subset by individual coach, data were confirmed to be nonparametric using the function shapiro test. Descriptive summary data are presented as median, minimum, and maximum values for each variable. The "rmcorr" package was used to run repeated-measures correlations on subjective measures to obtain Spearman rank correlation coefficients and 95% confidence intervals, which accounted for common within-individual association and nonnormal distribution.[11] Alpha was set at .05.

Results

Table 1 shows median, minimum, and maximum values for all variables. To provide insight into results across the 16 weeks, visual analog scale responses for mental fatigue, physical fatigue, and readiness to perform and sCort, sAA, and sAA:sCort are shown in Figures 1(A–C) and 1(D–F). Repeated-measures correlations (95% confidence interval) revealed a significant inverse relationship between mental fatigue and readiness to perform ($r = -.44$ [−.64 to −.17], $P = .002$) and a positive significant relationship between mental fatigue and physical fatigue ($r = .67$ [−.45 to .1], $P < .001$). Physical fatigue was not significantly related to readiness to perform ($r = -.19$ [−.45 to .1], $P = .208$). Figure 2 shows the rmcorr plot for each individual coach, where significance was detected, with separate lines fit to data reported by each coach.

Discussion

The present findings are the first to quantify elite coaches' experiences of mental fatigue. The individual variability in mental fatigue observed between coaches is in agreement with prior findings observed in athletes.[12] Coaches reported mental fatigue ratings equivalent to, or greater than, levels observed following interventions (eg, Stroop task) designed to induce mental fatigue and that have resulted in subsequent impairments in cognitive performance.[13] These elevations were observed despite the time point of assessment being before morning training (ie, minimal opportunity for adenosine to accumulate)[1] and following a period of allocated time away

Table 1 Weekly Individual Coach Mental Fatigue, Physical Fatigue, Readiness to Perform, sCort, sAA, and sAA:sCort During the 16-Week Preseason Phase

	Median (min–max)		
Variable	Coach 1	Coach 2	Coach 3
Mental fatigue (0 = none at all, 100 = maximal)	70.50 (25–86)	0.00 (0–51)	23.00 (15–76)
Physical fatigue (0 = none at all, 100 = maximal)	63.00 (28–85)	0.00 (0–20)	5.50 (0–17)
Readiness to perform (0 = not at all ready, 100 = maximally ready)	86.50 (61–100)	100.00 (23–100)	77.50 (41–94)
sCort, $nmol \cdot L^{-1}$	12.33 (8.42–17.31)	7.47 (4.20–9.70)	13.66 (6.81–19.66)
sAA, $\mu mol \cdot L^{-1}$	78.84 (52.40–113.06)	250.4 (158.80–307.20)	195.80. (86.55–495.85)
sAA:sCort	6.65 (3.20–12.80)	33.45 (21.10–61.70)	14.95 (4.90–35.50)

Abbreviations: sCort, salivary cortisol; sAA, salivary alpha-amylase.

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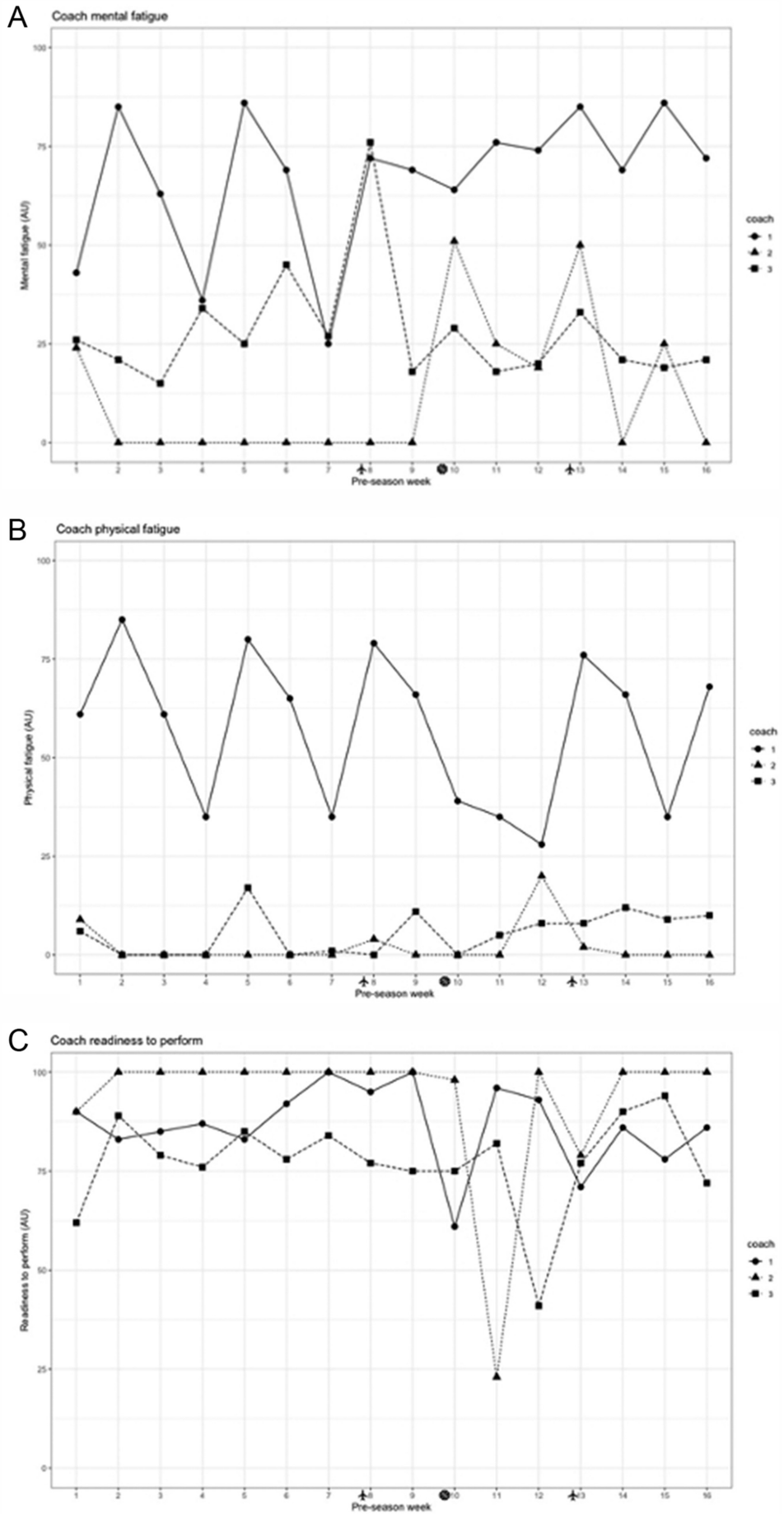

Figure 1(A-C) — Line graphs showing weekly individual coach (A) mental fatigue, (B) physical fatigue, and (C) readiness to perform during the 16-week preseason training phase. ✈ indicates travel for preseason match; ⊗, preseason tournament.

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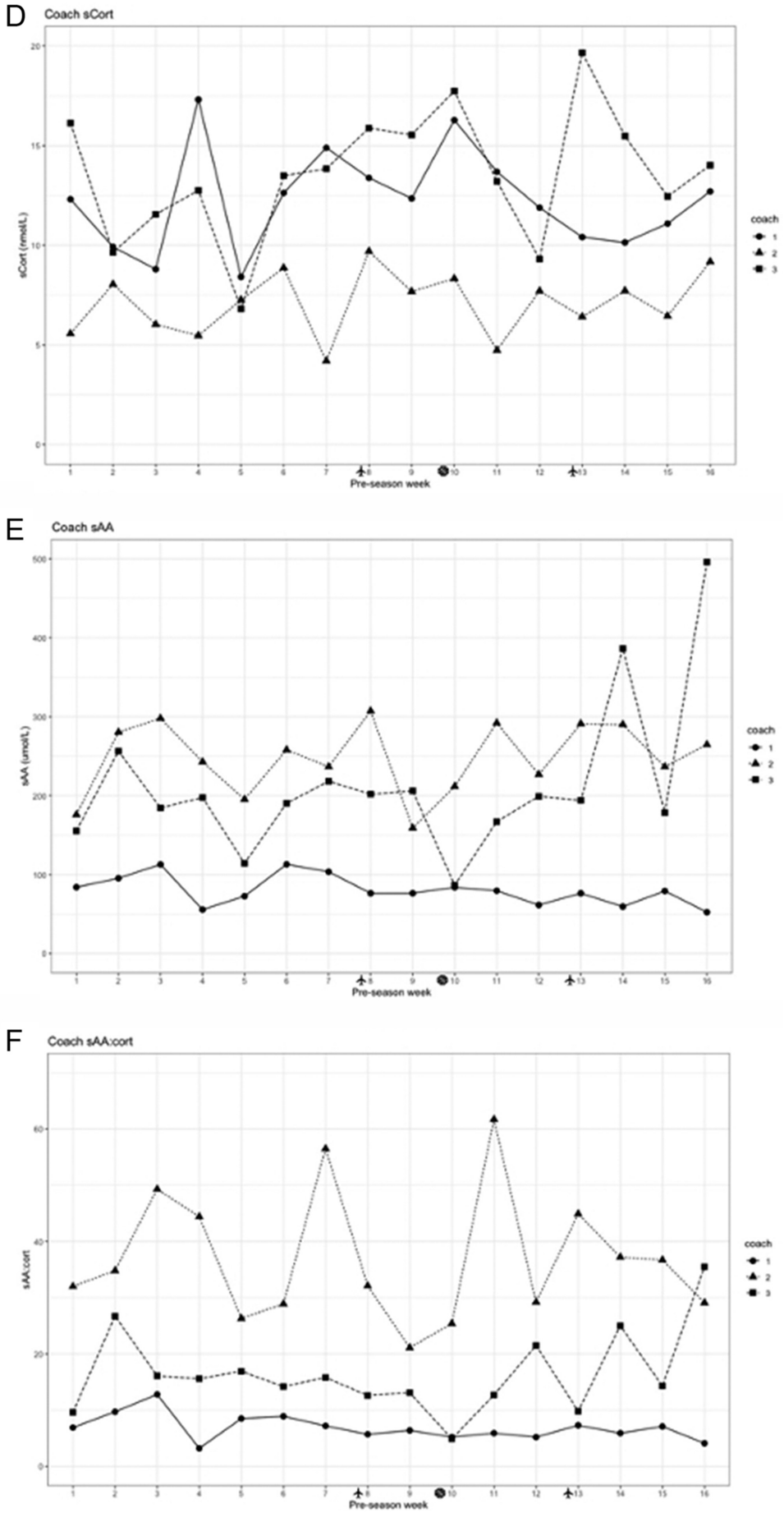

Figure 1(D-F) — Line graphs showing weekly individual coach (D) sCort, (E) sAA, and (F) sAA:sCort during the 16-week preseason training phase. ✈ indicates travel for preseason match; 🏀, preseason tournament; sCort, salivary cortisol; sAA, salivary alpha-amylase.

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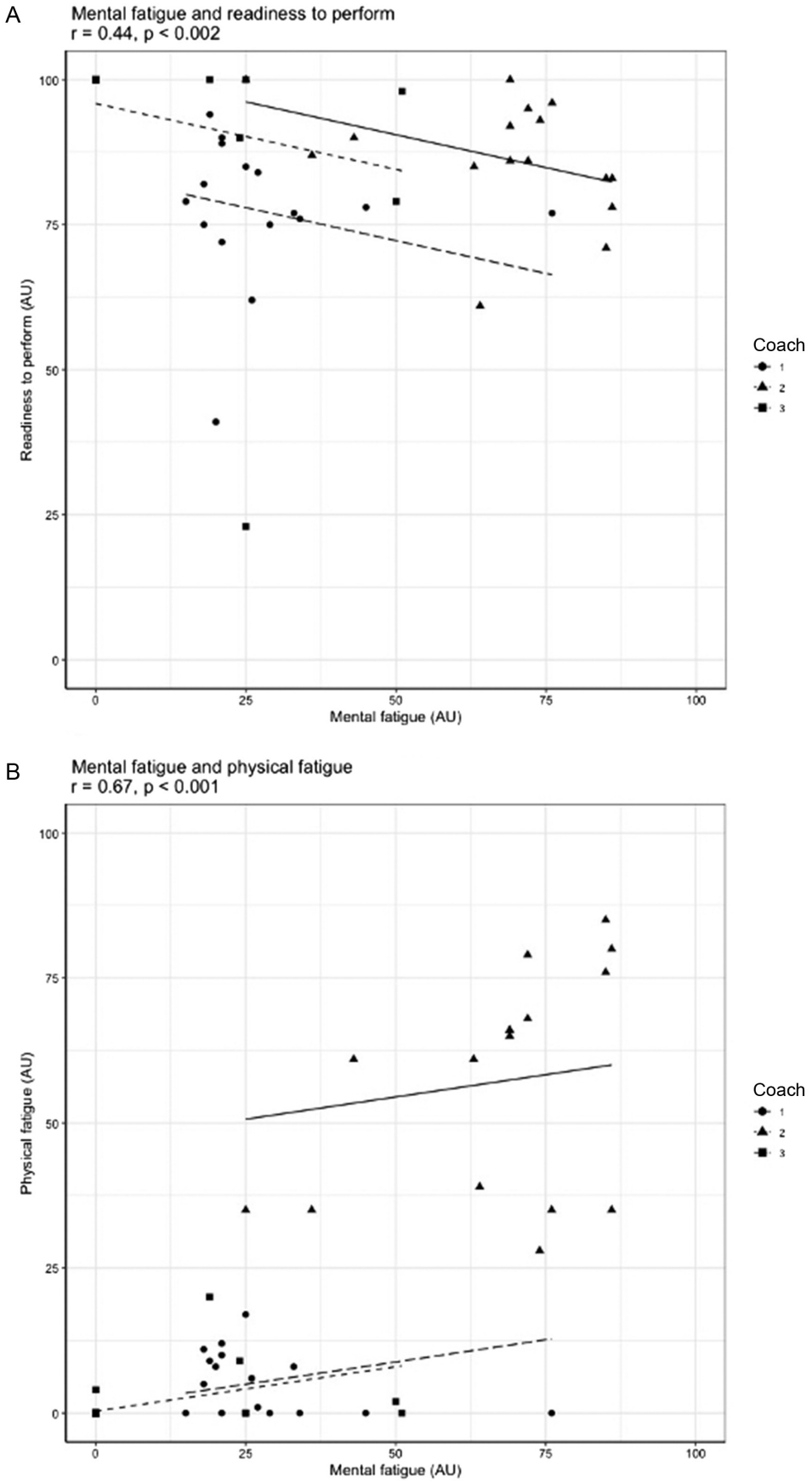

Figure 2 — Repeated-measures correlation plot showing significant individual coach relationships between variables of mental fatigue, readiness to perform, and physical fatigue.

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from required coaching duties. The observed sCort, sAA concentrations, and sAA:sCort also indicate coaches experienced psychophysiological stress via the hypothalamic–pituitary adrenal and sympathetic nervous activity during the preseason training phase.[10] For more context on the interpretation of sCort, sAA, and sAA:sCort results, please see Russell et al.[10]

Of particular interest is the contextual information surrounding time points where coaches reported elevated mental fatigue (Figure 1A). Elevated ratings were observed in weeks 8 and 12, immediately following weekends that required domestic interstate travel for "double-header" preseason trial matches. These time points required intensified coaching and selection decisions and game analysis and review in addition to air travel, changes in usual schedule, and time away from home. Such fluctuations and factors have previously been reported by athletes and staff as potential causes of mental fatigue in the elite sporting setting.[10,14] An intensified preseason tournament was held at the end of week 9, providing context for the subsequent elevated reports observed in week 10. The team coached by the participants recorded several losses during the tournament, anecdotally underperforming comparative with expectations. Negative match outcome has previously been associated with elevated subsequent mental fatigue in soccer players.[15] The present data indicate that this effect may also be observable in coaching staff, possibly due to increased emotional regulation, problem-solving demands, and potential upcoming season performance. Although coach-specific contextual factors that contribute to interindividual differences in mental fatigue are also of interest, further research is first needed to identify potential factors that explain individual susceptibility to mental fatigue.[16]

In line with previous research in athletes, a significant positive correlation, with 45% explained variance, was found between mental fatigue and physical fatigue.[10] A significant inverse correlation was also observed between mental fatigue and readiness to perform, with an explained variance of 19%. Accordingly, how coaches independently rate their mental fatigue appears to relate to how ready they feel they are to coach their athletes, with higher mental fatigue relating to lower readiness to perform. Further research, with a larger sample size, using subjective, behavioral, and (neuro)physiological markers and indicators of coaching performance, is necessary to investigate the potential cognitive impacts of elevated mental fatigue on subsequent coaching performance.

The consequences of the decisions and actions made by coaching and performance staff are likely to influence the athletes and, thus, their performance. Although likelihood and impact of mental fatigue may vary between coach type or performance support role, all coaches and staff are making decisions that will positively, or negatively, impact their athletes on a daily basis. Somewhat reflective of their contribution, substantial funding is invested into the selection and ongoing professional development of elite coaches and performance staff. A recent editorial highlighted that the performance of all humans can be impacted by mental fatigue and emphasized the risk for professionals employed in sport.[1] In consideration of the large investment made into staff, and the breadth of their impact on a sporting organization's assets, there is a need to understand mental fatigue and explore management or mitigation strategies to maximize the cognitive performance of coaching and performance staff. We recommend that future research considers the extent of mental fatigue across elite sporting contexts. Observational studies need to better understand and monitor the mental fatigue of coaches and other performance staff and the potential subsequent consequences on the quality of coach–athlete interactions. Furthermore, interventional and case study approaches should be used to build an evidence base of practical strategies to minimize mental fatigue and optimize cognitive performance of coaching and performance staff.

Practical Applications

Elite coaches and performance staff undertake cognitively demanding tasks daily and report elevated instances of mental fatigue. Educating staff on the presence and potential impacts of mental fatigue, in addition to strategies to manage or acutely mitigate mental fatigue, is recommended. Optimization of the cognitive performance of coaches and performance staff presents as a potential source of competitive advantage.

Conclusions

Coaches and performance staff experience individual fluctuations in, and instances of, elevated mental fatigue during an elite preseason training phase. Organizations and practitioners should work to be cognizant of coaches' mental fatigue and its potential influence on subsequent decisions, actions, and coach–athlete interactions.

Acknowledgments

We acknowledge and have great appreciation for the contribution of the practitioners to data collection and support of this research. Dr Marni Simpson is acknowledged for her assistance with the data collection process. Mr Gary Wilson, Dr Llion Roberts, and Ms Emma Harris are acknowledged for their support in the salivary analysis processes. Dr Suzanna Russell acknowledges the Australian Government Research Training Program and Queensland Academy of Sport—Sport Performance Innovation and Knowledge Excellence (Student Research Scholarship) for their support during the data collection phase of this manuscript. Dr Suzanna Russell was a recipient of a Sports Medicine Australia Research Foundation grant and expresses appreciation for the financial contribution made by Sports Medicine Australia to support the laboratory analysis of the project. Bart Roelands is a Collen-Francqui research professor and a member of the Strategic Research Program Exercise and the Brain in Health & Disease: The Added Value of Human-Centered Robotics (SRP17 and SRP77).

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